Seohyeong (Christine) Lee

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Research Interests

I am interested in discovering and synthesizing novel inorganic materials for next-generation energy applications.

- Non-equilibrium synthesis and phase engineering of solid-state inorganic materials (catalysts, battery applications)
- Exploratory synthesis for discovering novel functional chalcogenides and oxide phases
- Integration of high-throughput synthesis and machine-learning into experimental synthesis workflows

Education

Nanyang Technological University (NTU), *Singapore* — Aug 2021 – Jun 2025
Bachelor of Engineering in Materials Science and Engineering (Honours), Minor in Entrepreneurship
Advisor: Dr. Kedar Hippalgaonkar and Dr. Jose Recatala-Gomez
Final Year Project: Solid-State Bimetallic Catalyst Design Strategies for Light Olefin Production via CO_2 hydrogenation

Research Experience

Nanyang Technological University | Hip Lab — Aug 2025 – Current
Research Assistant (Advisor: Dr. Kedar Hippalgaonkar) — *Singapore*

- Project: Development of Fe–based thermal catalysts for selective CO_2 to C_5+ hydrocarbon conversion using Joule-heating synthesis *(NTU–A*STAR-UC Berkeley BEARS collaboration)*
- Optimized Joule-heating parameters (voltage, current, time) to achieve compositionally homogeneous catalysts.
- Characterized catalyst phase, morphology, and composition using XRD, SEM–EDS and ICP-OES.
- Collaborated with the Bayesian-optimisation sub-team to inform composition tuning and shorten the experimental cycle time.

Nanyang Technological University | Hip Lab — Jun 2024 – Jun 2025
Undergraduate Research Assistant (Advisor: Dr. Kedar Hippalgaonkar) — *Singapore*

- Project: AI-guided discovery and exploratory synthesis of low thermal conductivity materials
- Synthesized novel multinary compounds predicted by generative models for low lattice κ using solid-state and non-equilibrium routes.
- Performed phase and structural verification through XRD and SEM–EDS; analyzed cell parameters to compare with theoretical predictions.
- Collaborated with computational team to validate experimentally realized phases and refine synthesizability criteria.

Nanyang Technological University | Le Ferrand Research Group — May 2022 – Jan 2023
Undergraduate Research Assistant (Advisor: Dr. Hortense Le Ferrand) — *Singapore*

- Prepared ceramic slurries and optimized printing parameters, including G-code path modifications, to achieve uniform layer deposition and reduced surface defects.
- Assisted in magnetic alignment experiments to achieve controlled microstructures.

Publications

[1] **Effect of microplatelet orientation in 3D printed microplatelet-reinforced composites with bioinspired microstructures**
Weixiang Peng, Xin Ying Chan, Seo Hyeong Lee, Hortense Le Ferrand*
MRS Bulletin — 2024

[2] **Highly Selective CO_2 to C_5+ Hydrocarbons over Fe-based Catalysts via Direct Joule Heating**
Jose Recatala Gomez, Somya Verma, Seohyeong Lee, Moo Pei Ying, Delwin Tanto, Puneethkumar M. Srinivasappa, Andre KY Low, Joy Chun Qi Ng, Chuandayani Gunawan Gwie, Amol Amrute*, Kedar Hippalgaonkar*
(Manuscript in preparation)

[3] **Bayesian Optimisation-guided Discovery of Multimetallic Catalysts for Fischer-Tropsch Synthesis**
Jose Recatala Gomez, Somya Verma, Seohyeong Lee, Andre KY Low, Moo Pei Ying, Delwin Tanto, Puneethkumar M. Srinivasappa, Joy Chun Qi Ng, Chuandayani Gunawan Gwie, Amol Amrute*, Kedar Hippalgaonkar*
(Manuscript in preparation)

Presentations and Talks

AxCIS Lab – Thermocatalysis for CO_2 to Sustainable Aviation Fuel (SAF) Oct 2025
Annual Decarbonisation Scientific Symposium (Singapore)

Solvent-Free Joule Heating of FeCoNa Catalysts for CO_2 to C_5+ Hydrocarbon Conversion Oct 2025
11th Southeast Asia Collaborative Symposium on Energy Materials Malaysia

Solid-State Bimetallic Catalyst Design Strategies for Light Olefin Production via CO_2 hydrogenation May 2025
School of Materials Science and Engineering Undergraduate Research Presentation

Design and Build of an Affordable Laboratory Device using Additive Manufacturing Mar 2023
Society of Women Engineers Research Showcase

Work Experience

Infineon Technologies Jun 2023 – Dec 2023
Digital Transformation Intern *Singapore*

- Digitalized an Excel-based tracking system into a full-stack web application using HTML, CSS, and JavaScript, enabling engineers to monitor process milestones in real time.
- Automated data consolidation and reporting workflows, eliminating over 10 hours of manual work per week and minimizing human error.
- Integrated Tableau dashboards for KPI visualiation and improved data-driven decision-making process across multiple engineering teams.

Leadership & Service

ACS International Student Chapter, NTU Mar 2025 – Present
Deputy Secretary *Singapore*

- Coordinated with faculty advisors and ACS International representatives to plan and organize cross-institutional symposiums and outreach events.
- Led social media strategy and content design across Instagram and LinkedIn to enhance student engagement and chapter visibility.
- Represented ACS NTU Student Chapter at the 2025 SACSEM symposium.

Skills

Experimental: Joule heating synthesis, Exploratory synthesis, 3D Printing, high-energy ball milling
Characterization: XRD, Rietveld refinement, SEM, BET, ICP-OES
Data & Analysis: Python (Numpy, Pandas, Matplotlib), data visulization, LaTeX

References

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